

How the Internet Works

A concise technical overview for developers and students

Introduction

The internet is a global network of interconnected computers that communicate using a shared set of protocols. At its core, the internet is not a single entity but a collection of millions of private, public, academic, business, and government networks linked together. Data travels across this network in small units called packets, which are routed independently and reassembled at the destination.

The foundation of internet communication is the TCP/IP protocol suite — a set of rules that defines how data is formatted, addressed, transmitted, and received. Every device on the internet is assigned an IP address, which acts as its unique identifier, similar to a postal address.

Key Protocols

Several protocols work together to make internet communication possible. Each operates at a different layer and serves a specific purpose:

Protocol	Layer	Purpose
HTTP / HTTPS	Application	Transfer web pages and data securely
TCP	Transport	Reliable, ordered delivery of packets
UDP	Transport	Fast, connectionless data transmission
IP	Network	Addressing and routing of packets
DNS	Application	Translates domain names to IP addresses
TLS / SSL	Application	Encrypts data in transit

Table 1: Core internet protocols and their roles in the communication stack.

How a Request Travels the Network

When you type a URL into a browser, a precise sequence of events occurs in milliseconds. Understanding this sequence reveals how all the protocols above work together:

- 1. DNS Resolution:** Your browser asks a DNS resolver to convert the domain name (e.g. google.com) into an IP address. This lookup may be cached locally or resolved through a chain

of DNS servers.

2. TCP Handshake: Your device opens a TCP connection to the server's IP address via a three-way handshake: SYN → SYN-ACK → ACK. This establishes a reliable channel.

3. TLS Negotiation: For HTTPS sites, a TLS handshake follows, exchanging certificates and agreeing on encryption keys. All subsequent data is encrypted.

4. HTTP Request: Your browser sends an HTTP GET request over the encrypted connection, specifying the resource it needs (e.g. /index.html).

5. Server Response: The server processes the request and sends back a response — typically HTML, CSS, and JavaScript — broken into numbered packets.

6. Packet Routing: Each packet may travel a different physical path through routers worldwide, guided by BGP routing tables, before being reassembled by TCP at your end.

How the Internet Works

Continued — Infrastructure, Security, and the Web

Physical Infrastructure

The internet's physical layer is more tangible than most people realize. It consists of several types of infrastructure that carry data as light pulses, electrical signals, or radio waves:

Submarine cables: Over 400 undersea fiber-optic cables carry more than 95% of international internet traffic. A single modern cable can carry hundreds of terabits per second.

Internet Exchange Points (IXPs): Physical locations where different networks (ISPs, CDNs, cloud providers) interconnect and exchange traffic directly, reducing latency and cost.

Data centers: Large facilities housing servers and networking equipment. Major cloud providers operate dozens worldwide, enabling geographic redundancy.

Last-mile connections: The link between the ISP and your home — delivered via fiber, cable (DOCSIS), DSL over copper, or wireless technologies like 4G/5G.

The Domain Name System (DNS)

DNS is often called the 'phone book of the internet.' It is a hierarchical, distributed database that maps human-readable domain names to IP addresses. The hierarchy consists of root servers (13 logical groups worldwide), top-level domain (TLD) servers for .com, .org, .in, etc., and authoritative nameservers for individual domains. DNS responses are cached at every level — your OS, your router, your ISP — to reduce lookup time. The typical full resolution chain completes in under 50 milliseconds.

Security on the Internet

Because the internet is a shared, open infrastructure, security must be built into every layer. The key mechanisms are:

Mechanism	What it protects against
TLS/HTTPS	Eavesdropping and tampering on data in transit
DNSSEC	DNS spoofing and cache poisoning attacks
Firewalls	Unauthorized access to private networks

BGP Route Filtering	Route hijacking and traffic misdirection
Certificate Authorities	Impersonation of legitimate servers

Table 2: Common security mechanisms and the threats they mitigate.

The Web vs. the Internet

A common misconception is that the internet and the World Wide Web are the same thing. The internet is the underlying network infrastructure — cables, routers, IP addresses, and protocols. The Web is one application that runs on top of it, using HTTP/HTTPS to serve hyperlinked documents. Other applications that run on the internet include email (SMTP/IMAP), file transfer (FTP/SFTP), video streaming (RTMP/HLS), and peer-to-peer networks (BitTorrent).

Summary

The internet is a layered system. At the bottom, physical cables and wireless signals carry bits. IP provides a universal addressing scheme. TCP and UDP provide transport. DNS translates names to addresses. TLS provides encryption. HTTP delivers the web. Each layer depends only on the one below it, which is what makes the internet extensible — new applications can be built without changing the underlying infrastructure. This separation of concerns, known as the end-to-end principle, is the architectural idea that has allowed the internet to scale from a handful of research computers in 1969 to over five billion users today.

This document is intended as a RAG test corpus. Topics covered: TCP/IP, DNS, HTTP, TLS, physical infrastructure, internet security, and the web stack.